TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of Fabric Weight per Unit Area

Grams per square metre by the GSM cutter and by weighing the whole piece — two independent routes that must meet at the same number.

TE-3202 · QUALITY EVALUATION OF FABRIC AND APPAREL SESSIONAL · EXPERIMENT 2 · LEVEL 3 TERM 2

1. Aim and theory

Aim: to determine the mass per unit area of the reference fabric by (a) the small-specimen GSM cutter method and (b) weighing and measuring a whole piece, to show that both methods agree, and to convert the result between g/m² and oz/yd².

Mass per unit area — universally quoted as **GSM** (grams per square metre) — is the single most traded number in fabric commerce. It prices the fabric (yarn cost is bought by weight and sold by length), predicts drape, warmth and opacity, and is written into every specification and purchase order. It follows directly from construction: for a woven fabric it rises with thread density and yarn count, and falls if either is reduced — which is why GSM is also the fastest fraud and quality check in the mill: a fabric 5% light is almost always a fabric with threads or yarn substance missing.

$$\text{GSM (g/m}^2\text{)} = \text{mass of specimen (g)} / \text{area of specimen (m}^2\text{)}$$

1.1 The GSM cutter method

The GSM cutter (round sample cutter) cuts a disc of exactly 100 cm² = 0.01 m². The calculation then collapses to a multiplication:

$$\text{GSM} = \text{mass of 100 cm}^2\text{ disc (g)} \times 100$$

The method is fast and needs only a small sample, but it bets everything on a few square centimetres: the discs must be taken from spread positions, away from selvedges and creases, and at least four are averaged. The balance must read to 0.01 g, because 0.01 g on the disc is a full 1 g/m² in the result.

1.2 The whole-piece method

A full piece is measured for length and width (Experiment 1 methods) and weighed whole. Its area is the product of the two, and the same formula applies. This method averages the entire piece — no

sampling risk — but needs a piece that can be handled and a balance of suitable capacity, and the result includes selvedges. Agreement between the two methods is the teaching point of this experiment: sampling method and bulk method validating each other.

1.3 Unit conversion

Many specifications, especially from the United States, quote ounces per square yard. The exact conversion is:

$$\text{oz/yd}^2 = \text{g/m}^2 / 33.906 \quad \cdot \quad \text{g/m}^2 = \text{oz/yd}^2 \times 33.906$$

All weighing is done on fabric conditioned at 65% RH and 20 °C: cotton at 8.5% regain carries about 8 g of water in every 100 g weighed, so an unconditioned GSM is partly a weather report.

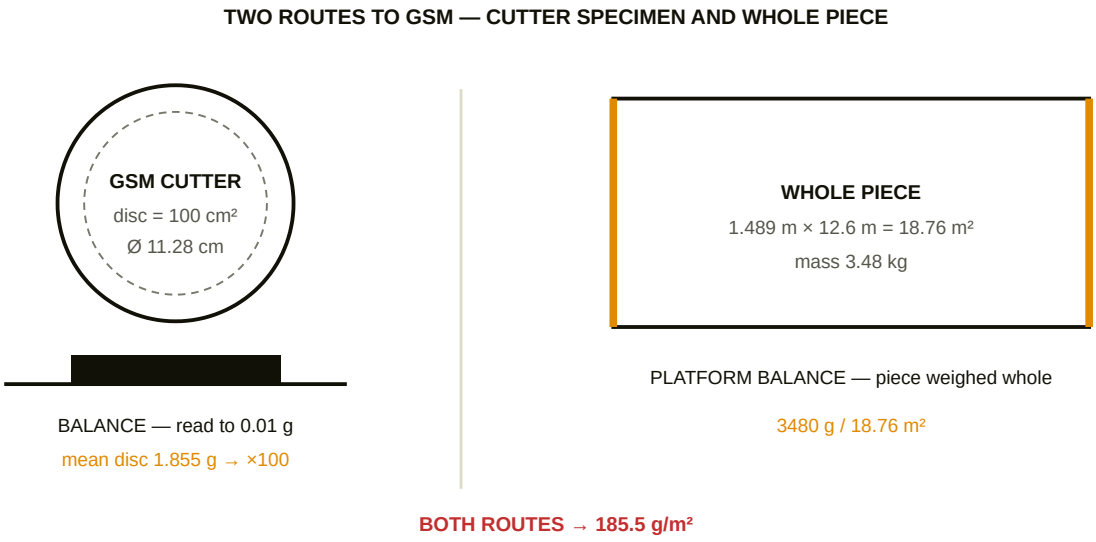


Figure 1 — Left: the GSM cutter cuts a 100 cm² disc whose mass in grams, multiplied by 100, is the GSM. Right: the whole-piece route divides total mass by measured area. Agreement between the two validates the sampling.

2. Apparatus and procedure

- **Apparatus:** GSM round sample cutter (100 cm²) with cutting mat and spare blades; analytical balance reading to 0.01 g; platform balance for the whole piece; measuring table, steel rule and tape; conditioned reference fabric (148.9 cm wide, Experiment 1).
- **Conditioning:** condition the fabric at 65% RH and 20 °C for at least 24 h and carry out all weighing in the same atmosphere.
- **Cutter method:** lay the fabric flat and smooth, place the cutter at spread positions at least 10 cm from any selvedge and clear of creases, press and rotate to cut four clean discs. Reject any disc with a ragged or incomplete edge. Weigh each disc to 0.01 g.

- **Whole-piece method:** measure the piece length (12.6 m) and width (1.489 m) by the Experiment 1 procedure, compute its area, then weigh the whole piece (3.48 kg) on the platform balance, taring any wrapping.
- **Calculation:** apply GSM = mass/area to both routes, average the cutter discs first, and convert the agreed result to oz/yd² by dividing by 33.906.

3. Worked calculation

Cutter method. Four discs weighed (g): 1.86, 1.83, 1.88, 1.85.

$$\text{Mean disc mass} = (1.86 + 1.83 + 1.88 + 1.85) / 4 = 7.42 / 4 = 1.855 \text{ g}$$

$$\text{GSM} = 1.855 \times 100 = 185.5 \text{ g/m}^2$$

The disc-to-disc range is 1.83–1.88 g, i.e. 183–188 g/m²: about ±1.4% about the mean, the normal local variation of a woven fabric — and the reason a single disc is never quoted alone.

Whole-piece method. Piece width 1.489 m, length 12.6 m, mass 3.48 kg = 3,480 g.

$$\text{Area} = 1.489 \times 12.6 = 18.7614 \approx 18.76 \text{ m}^2$$

$$\text{GSM} = 3{,}480 / 18.76 = 185.5 \text{ g/m}^2$$

Conversion.

$$185.5 / 33.906 = 5.47 \text{ oz/yd}^2$$

METHOD	MEASUREMENT	GSM
GSM cutter (mean of 4)	1.855 g per 100 cm²	185.5 g/m²
Whole piece	3,480 g / 18.76 m²	185.5 g/m²
Specification value	—	185.5 g/m² = 5.47 oz/yd²

WHY THE AGREEMENT MATTERS

Two methods with opposite weaknesses — the cutter samples only 400 cm² in total but weighs to 0.01 g; the piece method averages 18.76 m² but depends on two long measurements — return the identical 185.5 g/m². That cross-check is what licenses 185.5 as the GSM of the reference fabric for Experiments 5 and 7, where it enters the porosity and abrasion calculations.

4. Precautions

- Cut discs well away from selvedges: selvedge regions are denser and heavier and would bias the mean upward.
- Keep cutter blades sharp and the mat sound; a torn or stretched disc is not 100 cm², and the error passes straight into the GSM.
- Weigh promptly after cutting, in the conditioned room; small discs exchange moisture with room air within minutes.
- Do not stretch or iron the fabric before cutting — both change the area that the fixed cutter samples.
- For the whole-piece method, tare the wrapping and support the piece so no part drags on the floor or rests on the operator.
- Report the number of specimens and the spread, not just the mean: a mean of 185.5 from discs ranging 183–188 means something different from the same mean with a range of 170–200.

Practice (viva preparation)

1. Four GSM discs weigh 1.86, 1.83, 1.88 and 1.85 g. Find the GSM.
2. A piece 1.489 m wide and 12.6 m long weighs 3.48 kg. Find its GSM and compare with Question 1.
3. Convert 185.5 g/m² to oz/yd², showing the factor used.
4. Why does the balance for GSM work need to read to 0.01 g?
5. A fabric comes out 6% lighter than its specified GSM. What construction faults should be suspected first?
6. Why must GSM specimens be conditioned before weighing?
7. State one strength and one weakness of each method used in this experiment.

Answer key

1. Mean = 7.42 / 4 = 1.855 g per 100 cm² disc. GSM = 1.855 × 100 = 185.5 g/m².
2. Area = 1.489 × 12.6 = 18.76 m². GSM = 3,480 / 18.76 = 185.5 g/m² — identical to the cutter result, so sampling and bulk methods validate each other and 185.5 g/m² is accepted as the fabric's mass per unit area.
3. oz/yd² = 185.5 / 33.906 = 5.47 oz/yd². The divisor 33.906 converts grams per square metre to ounces per square yard (1 oz = 28.35 g and 1 yd² = 0.8361 m² combined).
4. Because the disc is only 0.01 m², the multiplication is ×100: every 0.01 g of weighing error or rounding becomes 1 g/m² in the reported GSM. A balance reading only to 0.1 g would leave the GSM uncertain by ±10 g/m² — useless against a typical ±5% tolerance.
5. Missing substance: fewer threads than specified (low EPI or PPI — check against the 70 × 59 construction), finer yarns than specified (higher Ne than ordered), or excessive size/finish removed

in processing without compensation. GSM is the sum of warp mass + weft mass + finish per unit area, so a shortfall must come from one of those terms; thread density is checked first because it is countable in minutes.

6. Cotton holds about 8.5% moisture at the standard atmosphere, and that water is weighed as if it were fabric. An unconditioned sample weighed on a humid day reads heavy and on a dry day reads light — by several g/m². Conditioning at 65% RH, 20 °C fixes the regain so GSM values are comparable.
7. Cutter method — strength: fast, precise weighing, needs only a small sample; weakness: sampling risk, a few hundred cm² must represent the whole piece. Whole-piece method — strength: averages the entire piece, no sampling risk; weakness: needs a handleable piece and a large-capacity balance, includes selvedges, and inherits any error in the long length and width measurements.
